

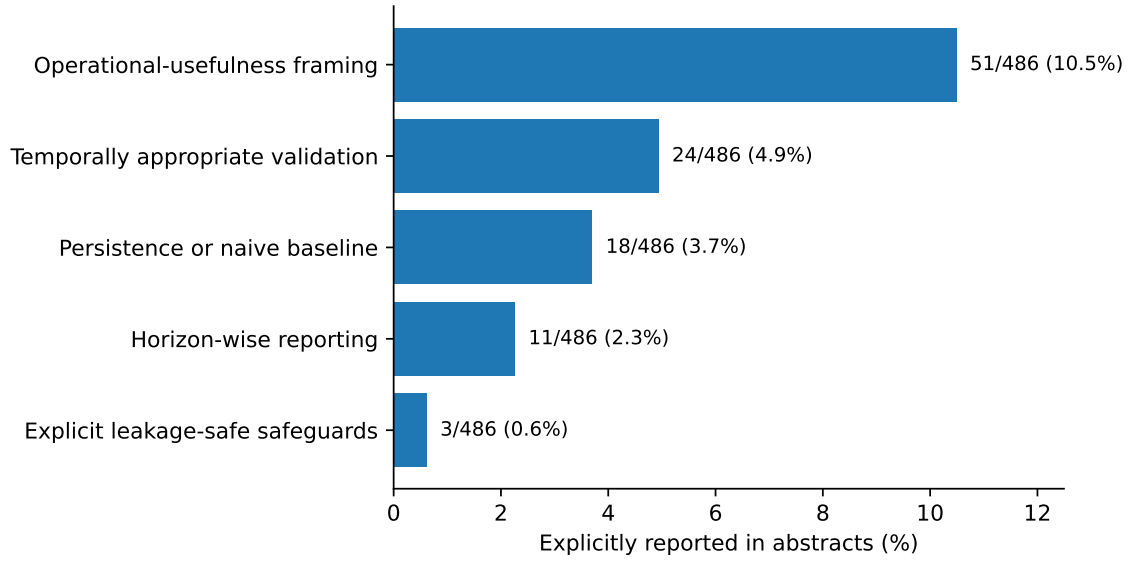


Figure 1: Abstract-level reporting visibility of key evaluation practices in particulate matter forecasting studies. Percentages are computed over 486 abstract-coded studies. Low percentages indicate reporting opacity, not confirmed absence of the corresponding practice in the full text.

Supplementary Material: Variance Retention Diagnostic for Daily PM10 Forecasts

S1. PRISMA reporting audit (supporting motivation)

Table 1: Abstract-level reporting visibility of evaluation-relevant practices in PM forecasting studies. Percentages are computed over 486 abstract-coded studies; low reporting rates indicate reporting opacity, not confirmed absence of the corresponding practice in full text.

Evaluation dimension	Explicitly reported	%	Interpretive function
Persistence or naive baseline	18/486	3.7	Minimum operational comparator
Horizon-wise reporting	11/486	2.3	Multi-step interpretability
Temporally appropriate validation	24/486	4.9	Deployment-like temporal credibility
Explicit leakage-safe safeguards	3/486	0.6	Protection against information bleed
Operational-usefulness framing	51/486	10.5	Decision relevance

S2. Monitoring stations

Table 2: Monitoring stations included in the study. Stations span industrial, traffic, urban background, suburban, and rural remote settings within the MITECO national air-quality network. Coverage refers to available daily PM₁₀ observations within the study period (2017–2024).

Station code	Name	Area type	Altitude (m a.s.l.)	Coverage (%)
ES0446A	Elx-Agroalimentari	Suburban/Industrial	44	80.4
ES1619A	Valencia-Vivers	Urban/Residential	11	91.7
ES0012R	Zarra-EMEP	Rural Remote EMEP	855	96.0
ES1870A	Barcelona El Port Vell	Suburban/Industrial	1	98.5
ES1418A	Alagon	Suburban/Traffic	235	97.5
ES1421A	Teruel	Urban/Background	915	96.7
ES0001R	San Pablo de los Montes	Rural	923	96.1
		Remote/Background		
ES1438A	Barcelona L'Eixample	Urban/Traffic	26	95.3
ES0567A	Barcelona Zona Universitaria	Urban/Background	24	95.2
ES1417A	Huesca	Urban/Traffic	488	95.0
ES0691A	Barcelona El Poblenou	Urban/Background	3	94.9
ES0559A	Barcelona Pl. Universitat	Urban/Traffic	24	94.9
ES1856A	Barcelona Vall d'Hebron	Urban/Background	136	94.5
ES1966A	Alcaniz Capuchinos	Urban/Industrial	340	94.3
ES2011A	Sant Vicenc dels Horts	Suburban/Industrial	65	94.1
	Alaba			
ES2091A	Alcanar Montecarlo	Rural	6	93.7
		Remote/Industrial		
ES2017A	Alcanar Llar de Jubilats	Rural	7	93.6
		Near-city/Industrial		